

Student's Habits and Academic Performance

An exploratory data analysis, followed by PCA and Clustering, to uncover key factors influencing student success. This presentation explores the intricate relationship between various student habits and their academic outcomes.

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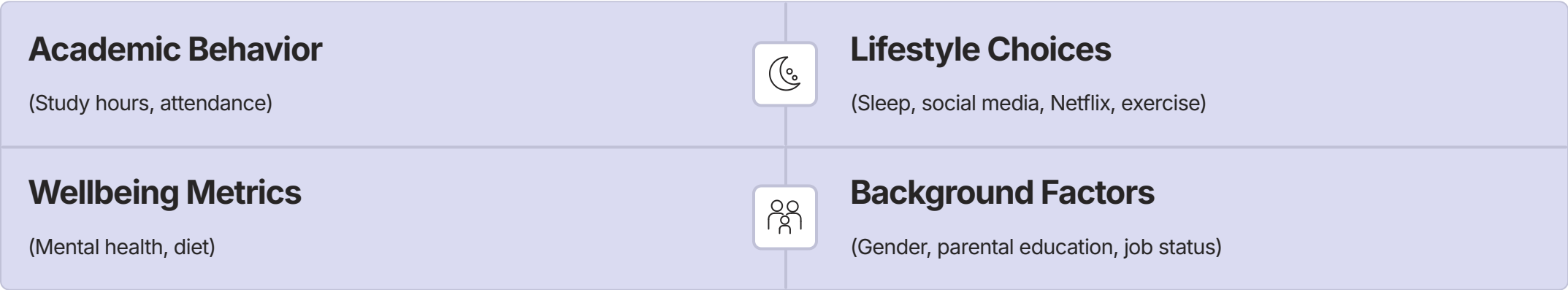


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Dataset Overview: A Glimpse into Student Life

- 1k anonymized university students
- Mix of numerical and categorical variables
- Comprehensive data spanning multiple domains

Key Data Domains



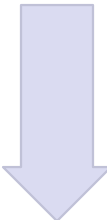
The ultimate goal: understand predictors for the [exam_score](#) target variable.

Data Cleaning & Preparation: Ensuring Reliability



Minimal Missing Values

Numerical variables showed **no missing values**, ensuring data integrity from the start.



Categorical Imputation

Missing entries in `diet_quality` and `parental_education_level` were strategically handled.



"Unknown" Category Strategy

Filled missing categorical values with **"Unknown"** to preserve information without distorting the dataset.



Duplicate Removal

Identified and removed duplicate rows to guarantee the uniqueness and accuracy of each student record.



Clean & Reliable Dataset

The rigorous cleaning process yielded a dataset that is robust, consistent, and ready for advanced analysis.

Why the "Unknown" Imputation Strategy?

Avoiding Data Distortion

Unlike mean/median imputation for numerical data, "Unknown" avoids fabricating values that could skew categorical distributions.

Interpretability & Clarity

This approach ensures that missingness is treated as a distinct category, providing meaningful context rather than an arbitrary fill.

Preserving Sample Size

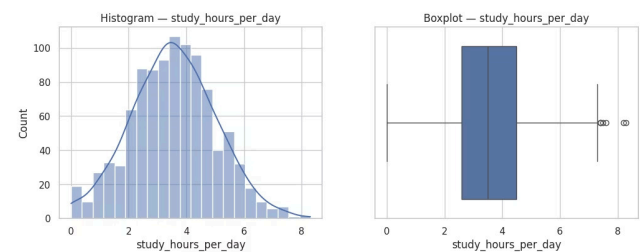
No rows were dropped, thus maintaining the full breadth of the dataset and preventing loss of valuable student information.

Non-Disruptive for Analysis

The "Unknown" category does not negatively impact subsequent PCA and clustering, which can treat it as another valid data point.

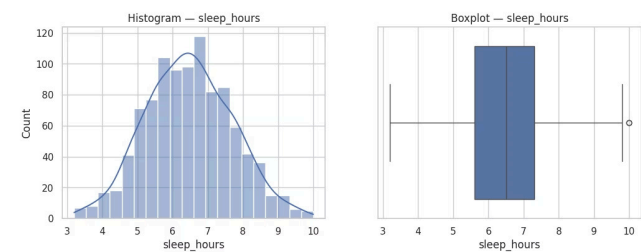
📌 This strategy is crucial for maintaining the integrity of categorical variables, offering a transparent and robust approach to missing data.

Univariate Analysis: Numerical Variable Insights



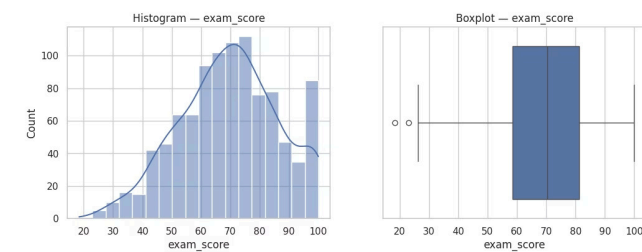
Study Hours

Exhibited **wide variability**, from minimal engagement to extensive study schedules.



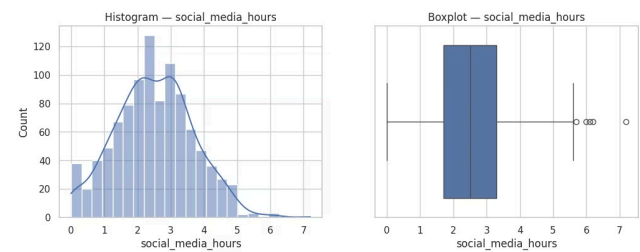
Sleep Hours

Also showed significant variation, reflecting diverse student sleep patterns.



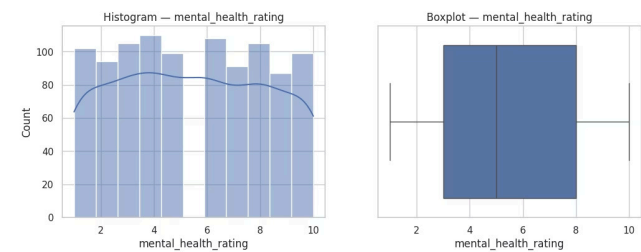
Exam Scores

Displayed a broad range, indicating a diverse academic performance spectrum.



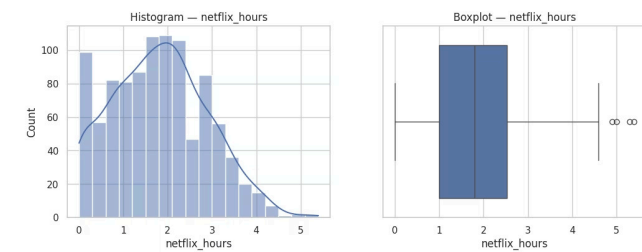
Social Media

Right-skewed, suggesting many students use these moderately.



Mental Health Ratings

Predominantly in the **moderate to high** range, which is a positive indicator.



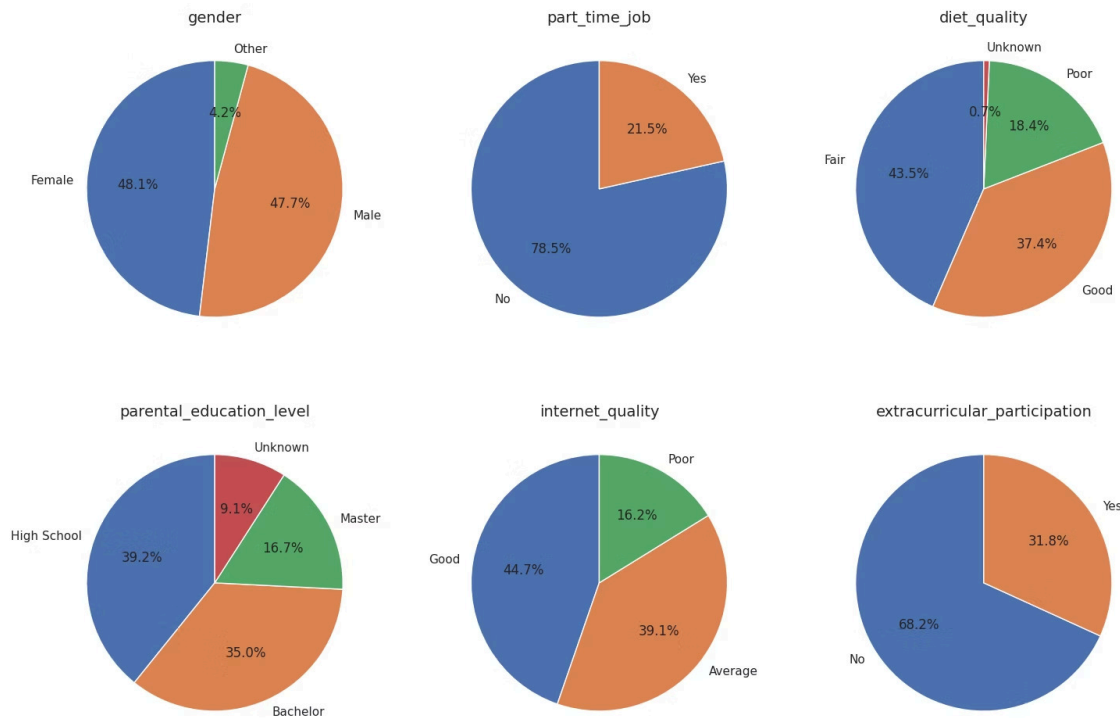
Netflix Hours

Right-skewed, suggesting many students use these moderately, but a few engage heavily more than other social media

Outliers, while present, were deemed realistic reflections of actual student behaviors rather than data errors.

Univariate Analysis: Categorical Variable Insights

Examining categorical variables uncovers underlying demographic and lifestyle distributions among the student population.



1 Diet Quality

A majority of students reported their diet quality as "Good" or "Fair", suggesting a moderate level of health consciousness.

2 Internet Quality

Primarily rated as "Average," indicating potential variations in access or connection reliability across students.

3 Job & Extracurriculars

Participation in part-time jobs and extracurricular activities was **imbalanced**, with fewer students engaged in these.

4 Gender

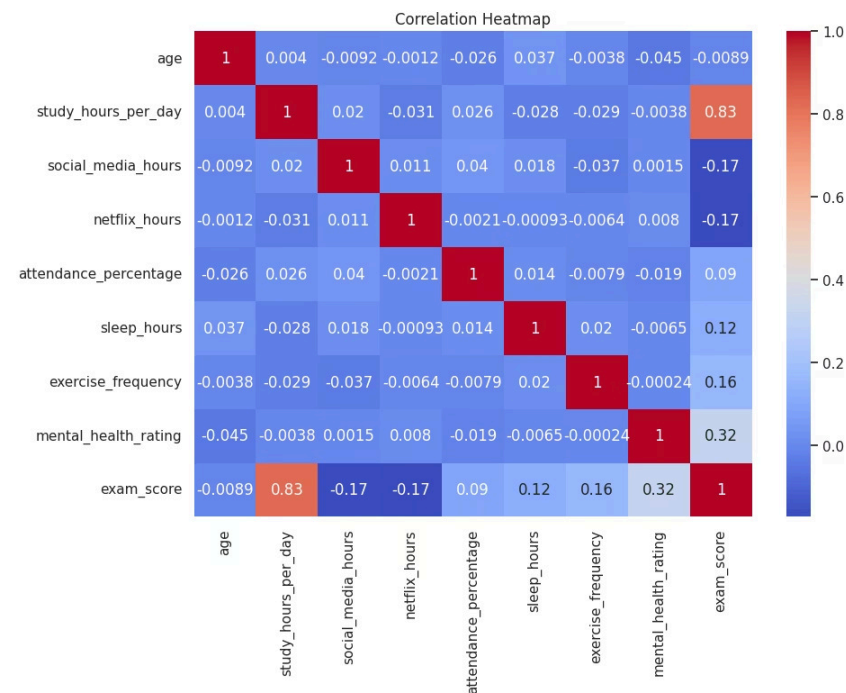
The gender distribution within the dataset was found to be **relatively balanced**, providing a representative sample.

Navigating Data Complexity: Bivariate vs. Multivariate vs. Univariate Analysis

Analysis Type	Variables	Purpose	Examples
Univariate	1	Understand distribution & spread	Histograms, boxplots, frequency tables
Bivariate	2	Study pairwise relationships	Scatterplots, boxplots, chi-square
Multivariate	3+	Discover complex patterns	PCA, clustering, correlation matrix

Bivariate Relationships with Exam Score

Understanding how individual factors influence academic performance.



Study Hours: The Strongest Link

A **strong positive correlation** ($r \approx 0.83$) indicates that more study hours lead to significantly higher exam scores.

Wellbeing: Moderate Influence

Mental health ($r \approx 0.32$), exercise, and sleep show a moderate positive relationship with exam performance.

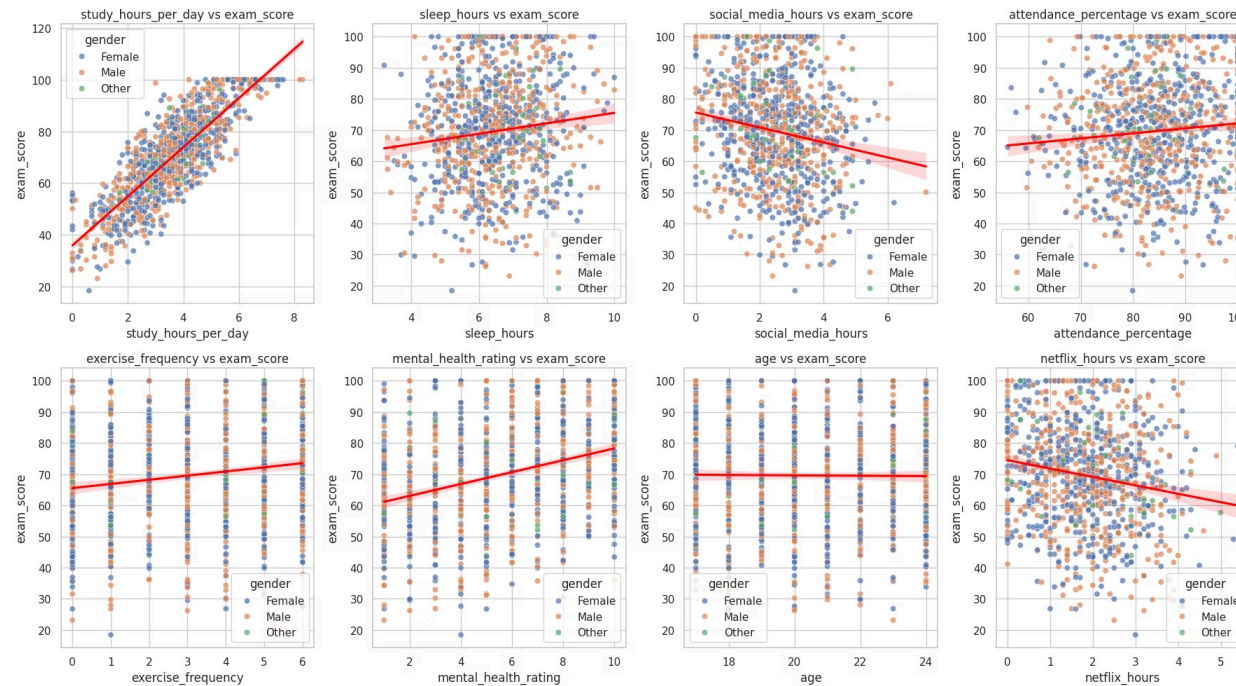
Digital Distractions: Negative Impact

Increased **social media and Netflix hours** are associated with lower exam scores.

Age: Minimal Effect

Age exhibits almost **no discernible relationship** with exam performance, suggesting it's not a primary predictor.

Bivariate Relationships



- **Study Hours: Strongest Predictor**

A robust positive linear relationship: students with more consistent study hours achieve higher scores. This the most influential factor.

- **Sleep Hours: Moderate Impact**

Moderate positive correlation: adequate sleep associated with improved academic behavior, contributing better focus & retention.

- **Digital Distractions: -ve Influence**

Clear negative relationships increased social media & Netflix consumption tends to correspond with lower exam scores.

- **Wellbeing & Attendance: Supportive Factors**

Exercise frequency and mental health ratings show positive trends, indicating a supportive role in performance. Attendance also has a weak but positive association.

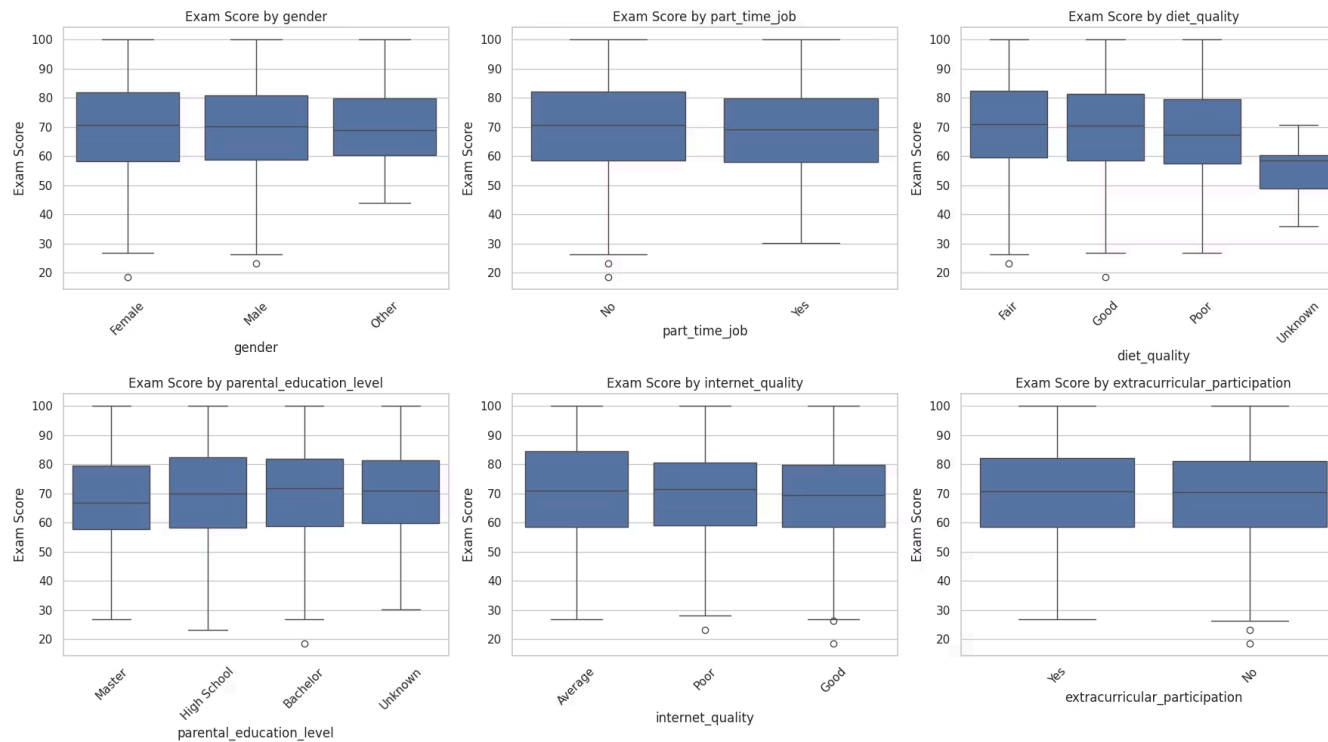
- **Age & Gender: Minimal Effect**

Age exhibits no relationship with exam performance.

Overall, academic performance is primarily driven by study behavior, supported by adequate sleep and good mental health, while excessive screen time negatively impacts results.

Categorical Variables vs. Exam Score

Exploring the impact of lifestyle and background on academic outcomes.



Diet Quality

Boxplots reveal a clear trend: students with a **better diet quality** generally achieve higher exam scores.

Parental Education

A **slight upward trend** in exam scores is observed with higher parental education levels.

Gender & Part-time Job

No strong differences were noted in exam scores based on gender or part-time job participation.

While less impactful than study-related behaviors, these categorical factors still offer valuable context.

Chi-Square & Categorical Associations

The Chi-square test of independence explores whether two categorical variables are statistically related or independent.



Chi-square (χ^2)

Measures how observed counts differ from expected, assuming independence. Small χ^2 suggests independence.



Degrees of Freedom (df)

Calculated as $(\text{rows}-1) \times (\text{columns}-1)$, indicating the complexity of the contingency table and more categories lead to larger df. Also, df helps determine the **shape of the Chi-square distribution**



p-value

Tests if these variables truly statistical significance: $p \leq 0.05$ implies a significant relationship.



Cramér's V

Measures association strength: values < 0.10 indicate a very weak relationship.

All tested categorical variable pairs (e.g., Gender vs. Part-time Job) had p-values > 0.05 and Cramér's V < 0.10 .

This indicates that categorical variables in the dataset are largely independent, confirming that academic performance patterns are primarily driven by numerical behavioral factors like study hours, sleep duration, and mental health, rather than redundant categorical characteristics.

PCA: Purpose & Preprocessing

Principal Component Analysis (PCA) is a powerful technique for reducing dimensionality and uncovering latent patterns within complex datasets.

01

Goal: Uncover Latent Patterns

PCA aims to transform variables into a new set of uncorrelated components, highlighting the most significant sources of variance.

02

StandardScaler for Numerical Data

Numerical features like 'Study Hours' and 'Exam Scores' were scaled to a standard normal distribution (mean 0, variance 1) to prevent features with larger scales from dominating the analysis.

03

One-Hot Encoding for Categorical Data

Categorical variables, such as 'Diet Quality' and 'Internet Quality', were converted into a binary format suitable for PCA, treating each category as an independent feature.

04

Dropped 'student_id'

The 'student_id' column was removed as it serves purely as an identifier and contributes no analytical information to the PCA model.

05

PCA Applied on Processed Matrix

Finally, PCA was applied to the fully preprocessed feature matrix, ensuring all variables contributed equally and meaningfully to the component extraction.

PCA Results & Interpretation

The initial principal components reveal significant underlying patterns, capturing a substantial portion of the dataset's variance.

First Components: Explaining Variance

The initial principal components collectively account for **over 80%** of the total variance in the dataset, indicating high explanatory power.

PC1: Academic Engagement

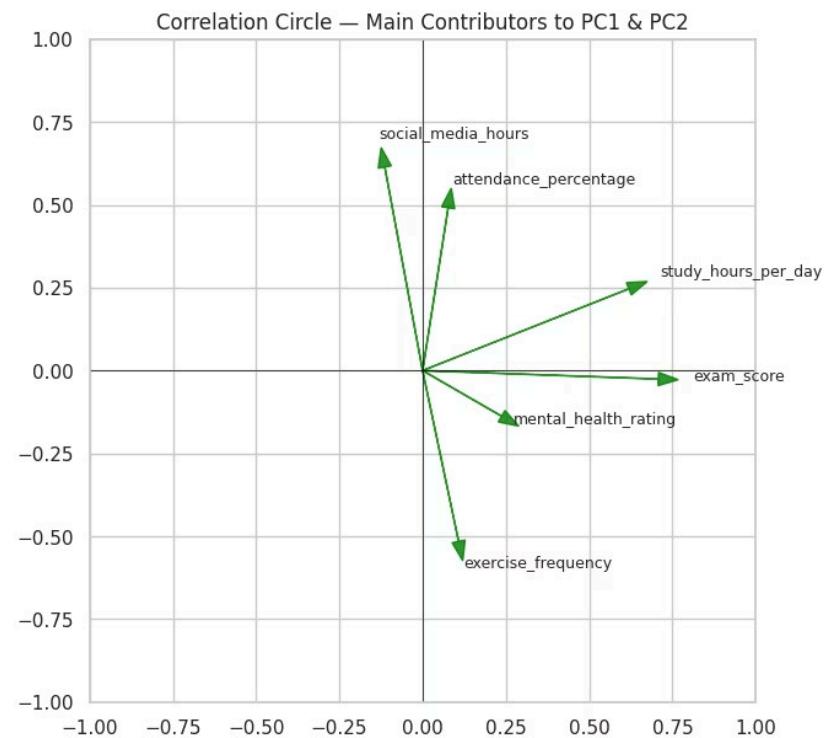
This component shows high positive loadings for **'Exam Score'** and **'Study Hours'**, highlighting a strong correlation with academic effort and outcomes.

PC2: Lifestyle Balance

Identified by strong loadings for **'Social Media Hours'**, **'Exercise Frequency'**, and **'Attendance'**, representing non-academic factors influencing student life.

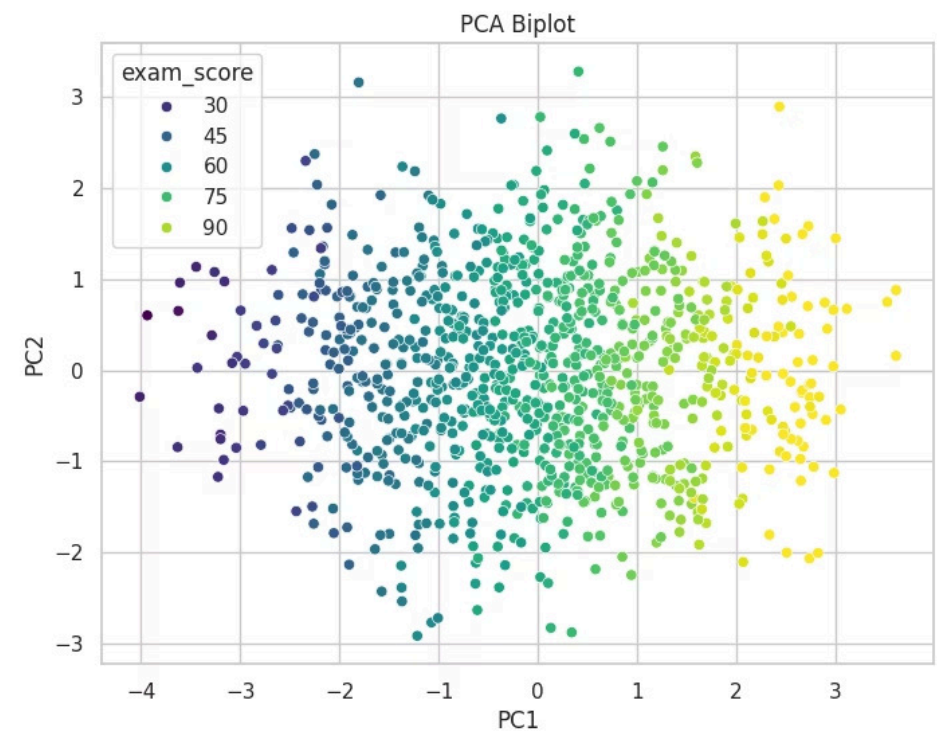
PCA Visualizations

Exploring the visual representation of Principal Component Analysis results to better understand student data structure.



Correlation Circle Insights

The correlation circle clearly delineates which original variables drive PC1 and PC2, showcasing a distinct separation between academic engagement and lifestyle-related factors.

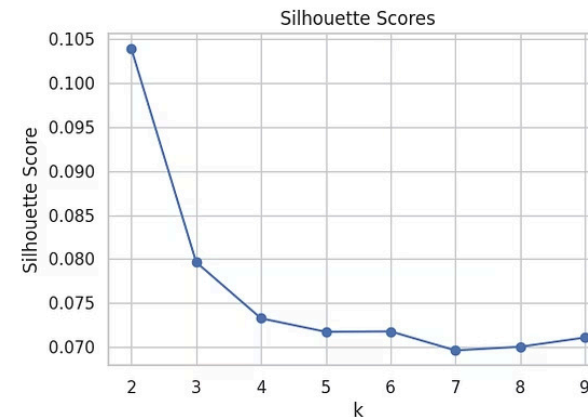
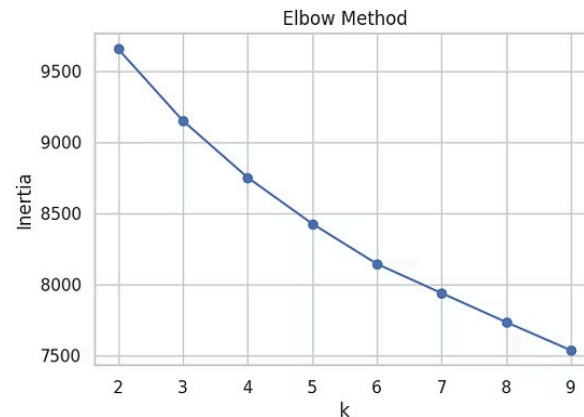


PCA Biplot Overview

The biplot illustrates how students are smoothly spread across the academic engagement spectrum. The exam score gradient aligns strongly with PC1, reinforcing its interpretation as an academic success dimension.

Clustering Analysis & Evaluation

Applying KMeans to identify distinct student segments based on their habits and academic performance and to utilize the insights gained from PCA to ensure meaningful clustering. from **k values 2 to 9**.



Elbow method (suggesting $k \approx 5$)

Inertia (Compactness only) = sum of squared dis of each point to its cluster centroid

High: Points are far from centroids, **Low:** Points are close to centroids.

So, More clusters = more centroids so At $k = n$, inertia = 0 (each point is its own cluster) so also, Inertia isn't enough alone to choose K.



Silhouette score (optimal at $k=2$) (Compactness + separation)

It measures **how well each point fits into its cluster** compared to others.

Close to +1: Well-clustered, **Around 0:** Overlapping clusters

Negative: Probably in the wrong cluster

So, High Silhouette score ensures points clearly belong to one cluster & clusters are meaningful & there's no unnecessary splitting

Final Choice: $K = 2$

Despite the Elbow method's suggestion, **$k=2$ was selected** due to its superior Silhouette score, indicating clearer and more distinct cluster separation.

Cluster Profiles & Final Insights

KMeans clustering revealed two distinct student profiles based on their study habits and academic performance.

CLUSTER 0

Lower Engagement

- Significantly **lower study hours** observed.
- Achieved **lower exam scores** (average ~56).
- Reported **lower mental health scores** and less frequent exercise.

CLUSTER 1

Higher Engagement

- Consistently engaged in **higher study hours**.
- Demonstrated **higher exam scores** (average ~83).
- Indicated **better mental health** and more regular exercise habits.

Overall Conclusions

Academic performance is primarily driven by study behavior and engagement.

Student **lifestyle and wellbeing** play a secondary but meaningful role in overall academic success.

PCA and clustering confirm clear and actionable **student performance profiles**, useful for targeted interventions.

ANOVA Validation of Results

ANOVA (Analysis of Variance) is a statistical method used to test whether there are significant differences between the means of three or more independent groups. In our analysis, it helps confirm the impact of various factors on student exam scores.

1

Between-Group Variance

Measures the differences in mean exam scores among different groups (e.g., different diet quality groups).

2

Within-Group Variance

Captures the natural variation in exam scores within each group, independent of the group factor.

3

F-statistic (strength of group separation)

A ratio of between-group to within-group variance, indicating the **strength of separation** between group means.

$F = \text{Variance within groups} / \text{Variance between groups}$

- **Large F** → groups are very different
- **Small F** → groups overlap heavily

4

p-value (statistical significance)

Indicates the **statistical significance** of the observed differences; a low p-value suggests the differences are not due to chance.

p value:

- **< 0.05** → Statistically significant difference
- **≥ 0.05** → No statistically significant difference
- **p ≈ 0** → overwhelming evidence

ANOVA Results: Exam Score by Variable

We applied ANOVA to assess how different variables and the derived KMeans clusters influence student exam scores.

Variable	F-Statistic	p-value	Interpretation
Diet Quality	2.73	0.042	statistically significant effect
Parental Education Level	0.65	0.58	No significant difference
Internet Quality	1.46	0.23	No significant difference
Cluster (KMeans)	1723.24	< 0.001	Very strong, decisive difference

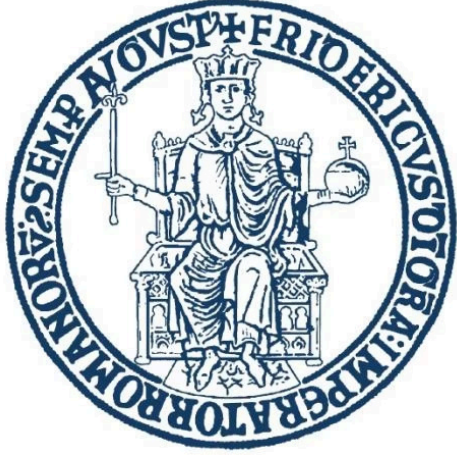
Key Insights

Lifestyle and background factors alone explain **limited variance** in exam scores.

Diet quality shows a **modest but statistically significant** effect on exam performance.

Clusters derived from PCA and KMeans exhibit **extremely large and significant differences** in exam scores.

This strongly validates that PCA + clustering effectively identifies **meaningful academic performance profiles**.



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**Any Questions,
Thanks.**